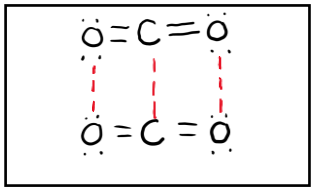
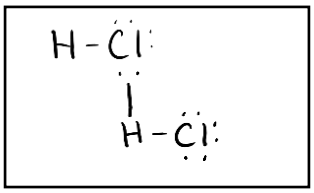
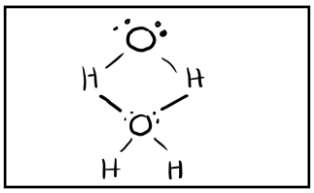
1. There are 2 types of forces that occur at the molecular level: **intra**molecular and **inter**molecular forces.
   a. An **intra**molecular force is considered a _____.
   b. How is an **intra**molecular force represented in a Lewis diagram?

   c. In the box to the right, complete the Lewis structure for HF.
   d. Draw an arrow that points to the **intra**molecular force.
   e. By this point, you should know that an **intra**molecular force occurs inside of a molecule. Briefly describe where an **inter**molecular force would occur.

H F

   f. Which diagram below correctly represents an intermolecular force?

   g. Why can 2 molecules be attracted to each other? What causes an IMF?

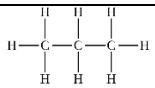
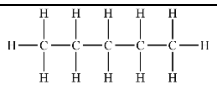
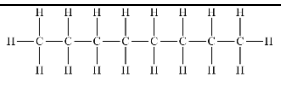
2. London dispersion forces (LDF) are the **weakest** intermolecular force of the IMFs we will discuss.
   a. Why are LDFs the weakest? Your answer should discuss the dipole that is formed.

   b. Discuss why all molecules that exist form LDFs.

   c. Which molecules will form **only** LDFs as their main IMF?

   d. The data table to the right shows multiple molecules that only have LDFs.
      i. What trend do you notice with the chemical formula/structure and the boiling point?

      ii. Each molecule only forms LDFs, explain what causes the trend you talked about in i.

Formula	Structure	State of Matter @ Room Temp	Boiling Point
C_3H_8		gas	-42°C
C_5H_{12}		liquid	36°C
C_8H_{18}		liquid	125°C

1. The table to the right shows 3 different molecules.

   a. Draw the Lewis structure, label as polar/non-polar, and determine the main IMF for each molecule in the table to the right.

   b. For each molecule, label each as having a temporary dipole or permanent dipole. Write it in the box provided to the right.

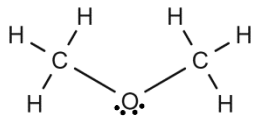
   c. In the box provided below, **draw 2 CH_3Cl** molecules and correctly show the attraction between the two molecules.

Formula	Lewis Structure	Polarity	IMF	Boiling Point
Water (H_2O) _____ dipole				100ºC
Methyl Chloride (CH_3Cl) _____ dipole				-24ºC
Methane (CH_4) _____ dipole				-161ºC

   d. Using the strength of the IMFs, briefly discuss the differences in boiling points represented above in the table.

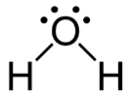
2. The rest of this page is dedicated to determining the IMFs that a molecule will form. For each molecule below, draw the Lewis structure, determine if it's polar or non-polar, and write out all IMFs it will form.

Carbon tetrachloride	Carbon monoxide	Carbon disulfide	H_2S
Polarity: _____ IMFs: _____	Polarity: _____ IMFs: _____	Polarity: _____ IMFs: _____	Polarity: _____ IMFs: _____

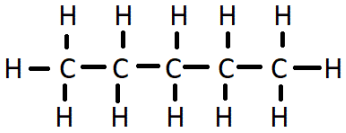
HF	Oxygen Gas	Diethyl Ether	HCN
			
Polarity: _____ IMFs: _____	Polarity: _____ IMFs: _____	Polarity: _____ IMFs: _____	Polarity: _____ IMFs: _____

1. The Lewis structure of 3 molecules have been provided below.

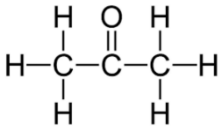
Water



Pentane



Acetone

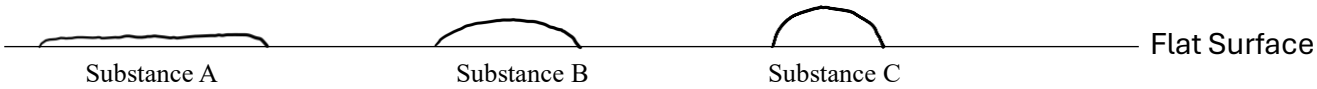


Main IMF: _____

Main IMF: _____

Main IMF: _____

a. Determine the main IMF that each molecule will form.
b. A student places 1 drop of each substance on a flat surface and draws what they see.



c. What property is this student testing? _____
d. Unfortunately, the student mixed up the substances and did not draw them in order. How can the student figure out which substance is which?

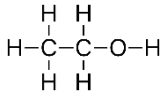
e. Based on your answer above, determine the identity of each substance.
   i. Substance A: _____
   ii. Substance B: _____
   iii. Substance C: _____

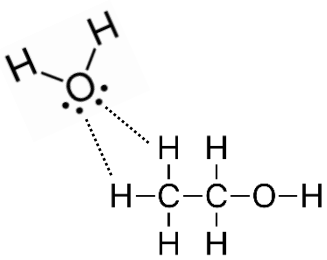
f. The student allows the drops to sit out for 10 minutes and notices that 1 of the substances has completely evaporated. Which substance do you expect that to be? Make sure your answer references the IMF present and the relative strength.

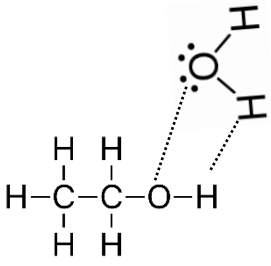
g. The data table to the right shows the boiling point for each substance above.
   i. Water has a much higher boiling point than pentane or acetone. In the box below, draw 2 water molecules and show how each is attracted to each other.

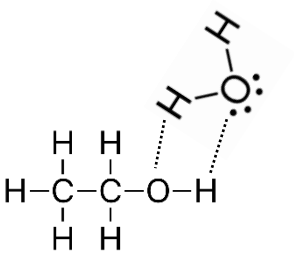
Water	100°C
Pentane	36°C
Acetone	56°C

h. The structure of ethanol is provided to the right. Ethanol can form hydrogen bonding with water molecules. Which diagram below correctly shows how ethanol hydrogen bonds with water.







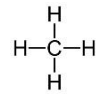
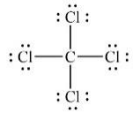


1. Place the molecules in order of increasing boiling point. (HF, O_2, CO_2)
   a. _____ < _____ < _____

2. Which substance will be the most volatile? (Highest evaporation rate)
   a. C_2H_6 b. H_2O c. CH_3OH

   Explain:

3. A student is comparing the boiling points of two compounds that are both non-polar. The Lewis structure and boiling points are provided in the table.

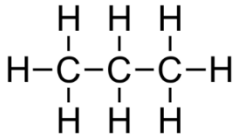
Lewis Structure	Boiling Point
	-162°C
	76°C

   a. What is the main IMF that is found in both molecules? _____
   b. Why is the boiling point for CCl_4 much larger than CH_4?

   c. The $\Delta H_{vaporization}$ for CCl_4 is 39.4 kJ/mol. Calculate the amount of energy, in kJ, that would be needed to vaporize 10.0 grams of CCl_4.

   d. Would you expect the $\Delta H_{vaporization}$ for CH_4 to be lower or higher than CCl_4? Explain your answer.

4. Propane, C_3H_8, is a colorless and odorless gas at room temperature. The structure of propane has been provided below.



   a. How many liters of propane are present in 7.50 grams of propane at STP?

   b. A student combusts the propane gas in the presence of oxygen gas. Write the full balanced reaction below.

   c. The enthalpy of the reaction, $\Delta H_{reaction}$, is measured to be -2200.0 kJ/mol rxn. How many kJ of energy are released if the 7.50 grams of propane are combusted?